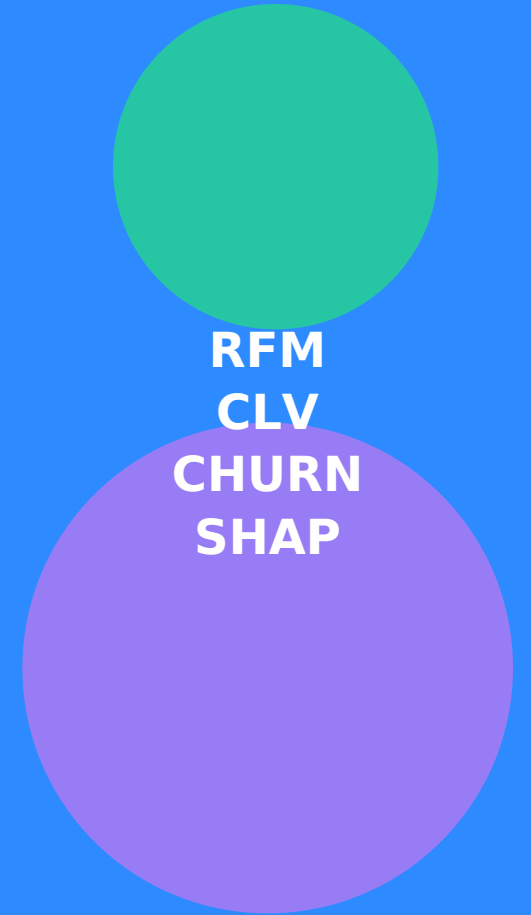


Customer Intelligence, CLV & Churn Prediction Platform

A production-style analytics portfolio connecting customer value, retention risk, explainability and next-best-action decisions.

Murat Miraç Gedik | Professional Portfolio Project | July 2026



Executive Decision Summary

Value, risk, retention and campaign economics in one view

CUSTOMERS

6,000

4,067 active in last 12 months

NET REVENUE 12M

TRY 55.6M

21.3% gross-margin rate

PREDICTED CLV

TRY 11.9M

Forward 12-month margin

HIGH / CRITICAL RISK

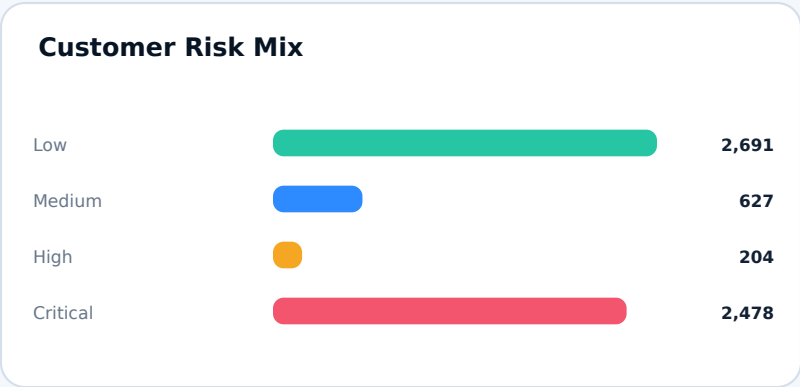
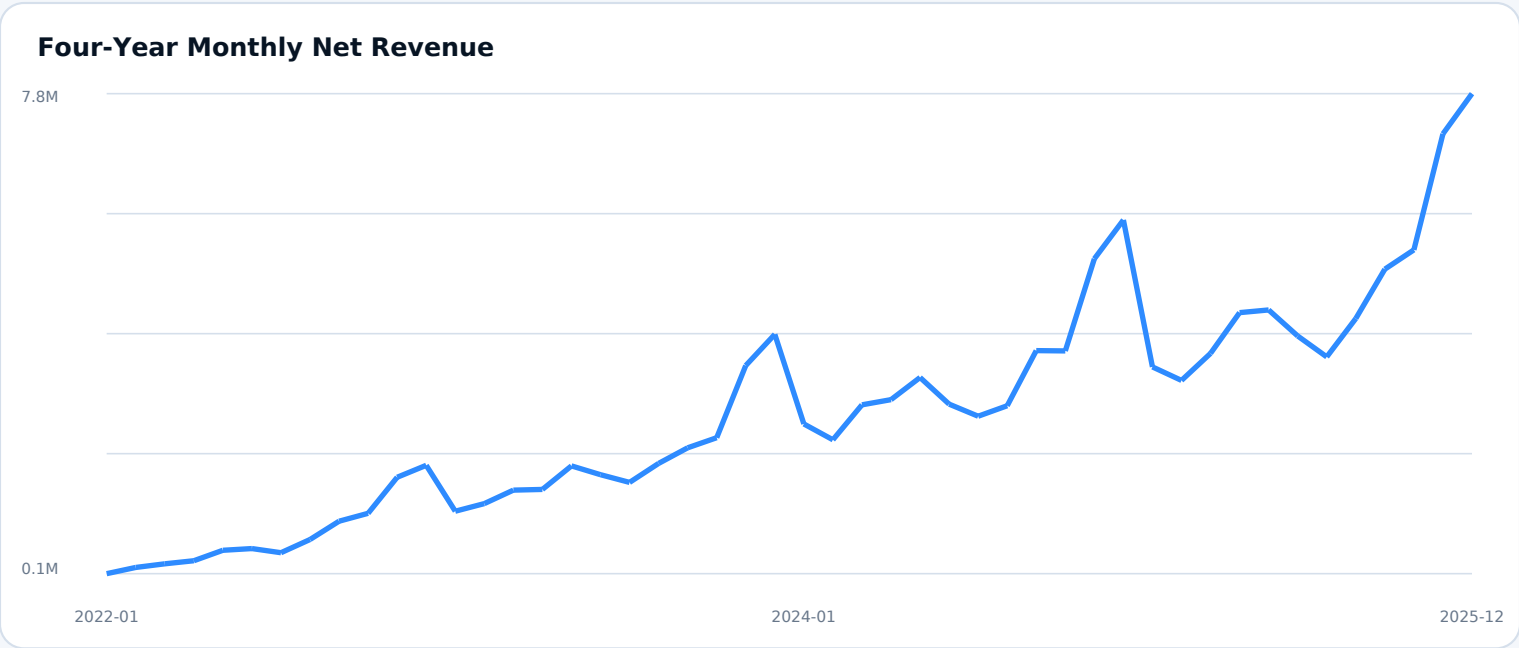
2,682

TRY 6.0M exposure

CAMPAIGN UPSIDE

TRY 547K

TRY 236K allocated budget



Decision message

2,682 customers require value-aware retention. The champion model produces 1.77x top-decile lift.

Business Problem & Decision Framework

From fragmented signals to controlled customer action

Fragmented view

Transactions, engagement, support and campaign signals are often disconnected.

Reactive churn

Interventions begin after value and relationship strength have already deteriorated.

Value-blind targeting

Risk alone can overfund low-value customers and miss profitable retention.

Opaque decisions

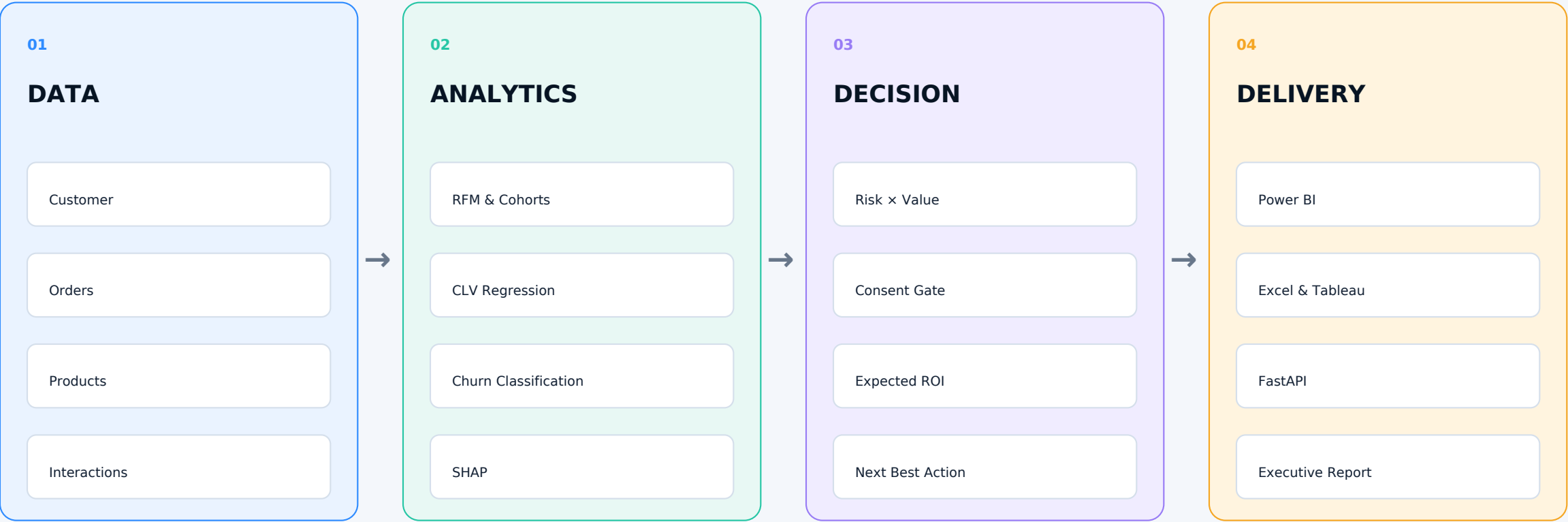
Unexplained scores reduce trust, adoption and governance quality.

Core decision

Which customer should receive which intervention, at what time and with what budget?

Solution Architecture

A reproducible path from source data to executive action

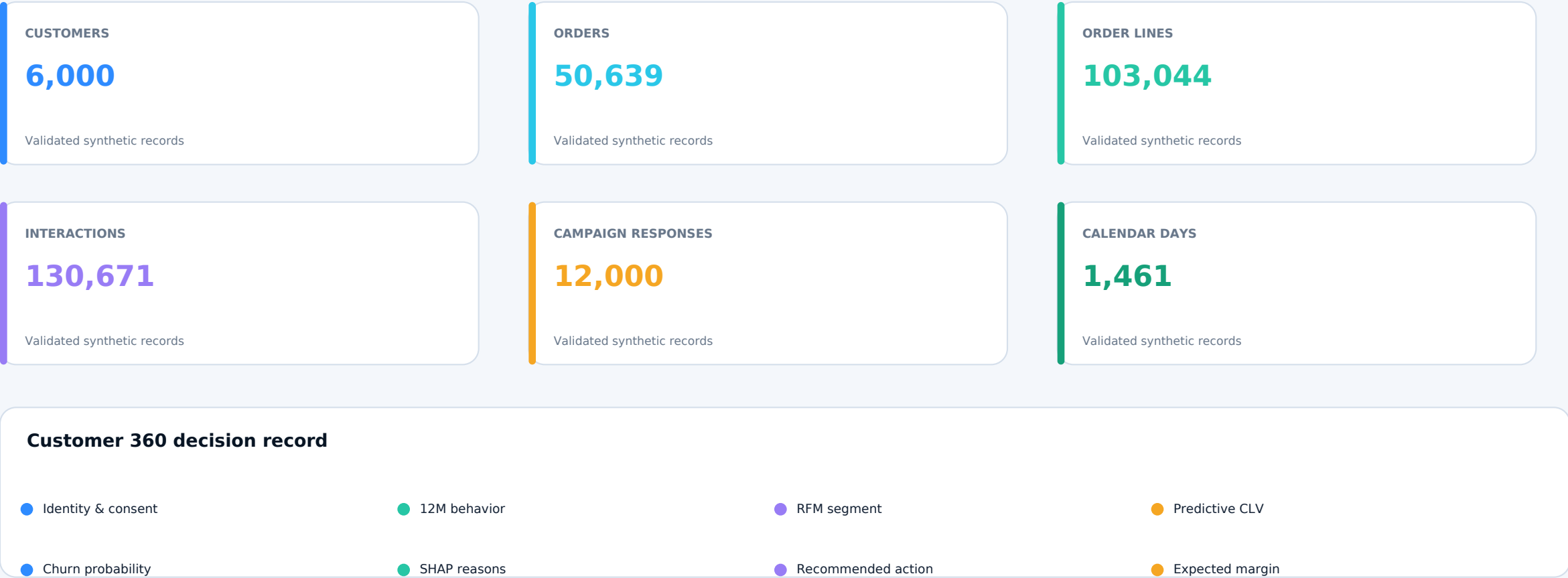


Operating principle

One customer key, one governed feature set and one validated decision record across every delivery channel.

Data Foundation & Customer 360

Four years of relational, privacy-safe synthetic portfolio data

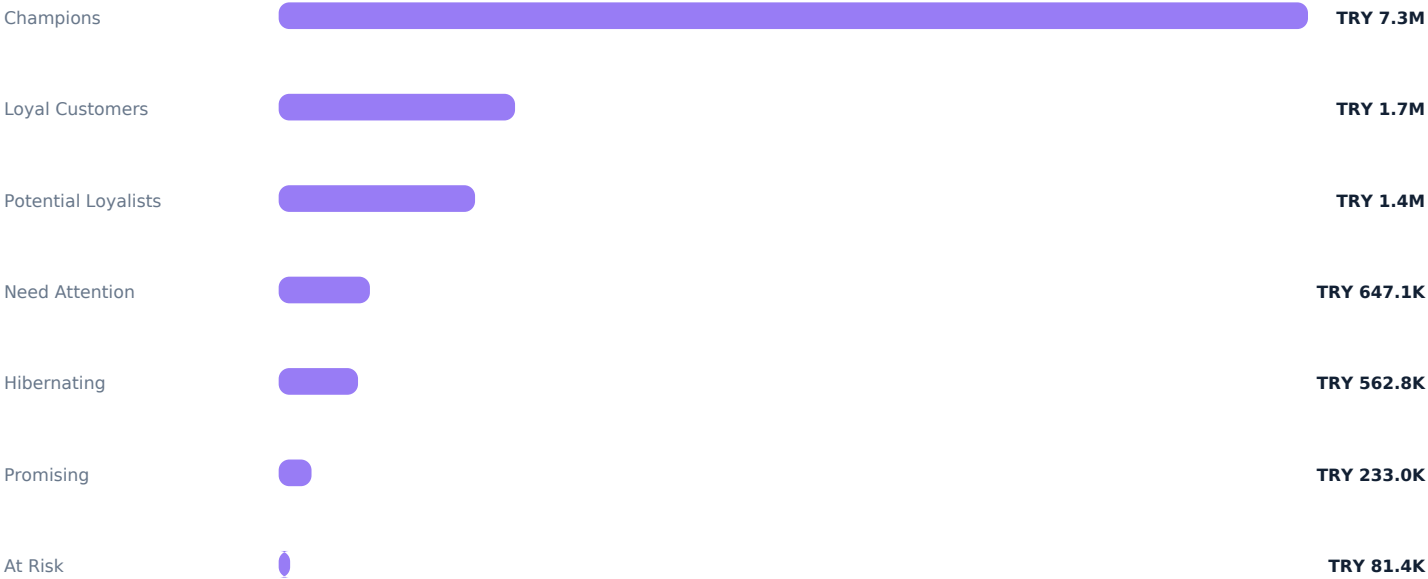


Sources: dim_customers.csv; fact_orders.csv; customer_360.csv; data_quality_report.csv

RFM Segmentation

Behavioral groups linked to portfolio value and retention risk

Predicted CLV by RFM Segment



Average Churn Risk



Commercial interpretation

Champions anchor the value pool. At-risk and hibernating groups require selective, economics-led treatment rather than blanket discounting.

Sources: rfm_scores.csv; segment_summary.csv

Cohort Retention

Acquisition quality measured beyond the first purchase

First Purchase Cohort x Months Since Acquisition

2024-07	100%	19%	42%	40%	59%	46%	31%	31%	33%	26%	26%	35%	29%
2024-08	100%	50%	35%	43%	50%	32%	33%	30%	44%	28%	35%	26%	24%
2024-09	100%	37%	52%	52%	28%	28%	31%	24%	31%	27%	26%	29%	27%
2024-10	100%	50%	46%	33%	23%	25%	32%	32%	23%	25%	27%	28%	32%
2024-11	100%	47%	33%	31%	36%	38%	41%	34%	31%	29%	35%	39%	42%
2024-12	100%	37%	27%	34%	36%	34%	27%	28%	30%	30%	31%	38%	43%
2025-01	100%	37%	30%	27%	31%	32%	28%	26%	31%	31%	36%	40%	
2025-02	100%	42%	40%	39%	39%	39%	37%	43%	34%	43%	42%		
2025-03	100%	45%	31%	27%	27%	36%	27%	36%	40%	48%			
2025-04	100%	37%	27%	30%	37%	37%	32%	43%	39%				
2025-05	100%	36%	38%	27%	39%	40%	44%	46%					
2025-06	100%	26%	29%	33%	37%	47%	47%						
2025-07	100%	37%	36%	31%	42%	48%							
2025-08	100%	42%	40%	49%	50%								
2025-09	100%	37%	42%	49%									
2025-10	100%	51%	50%										
2025-11	100%	54%											
2025-12	100%												

M3 RETENTION

35.1%

Average across cohorts

Management use

Compare acquisition months, identify early decay and connect cohort quality to channel, onboarding and offer strategy.

Sources: cohort_retention_matrix.csv; cohort_retention_long.csv

Predictive Customer Lifetime Value

Twelve-month gross-margin contribution for value-based prioritization

Model	MAE	RMSE	R ²	Spearman
Random Forest Regressor	TRY 1.3K	TRY 2.2K	0.564	0.792
Histogram Gradient Boosting Regressor	TRY 1.3K	TRY 2.2K	0.550	0.787
Ridge Regression	TRY 1.4K	TRY 2.4K	0.477	0.703

PORTFOLIO CLV

TRY 11.9M

Forward 12 months

CHAMPION R²

0.564

Time-based holdout

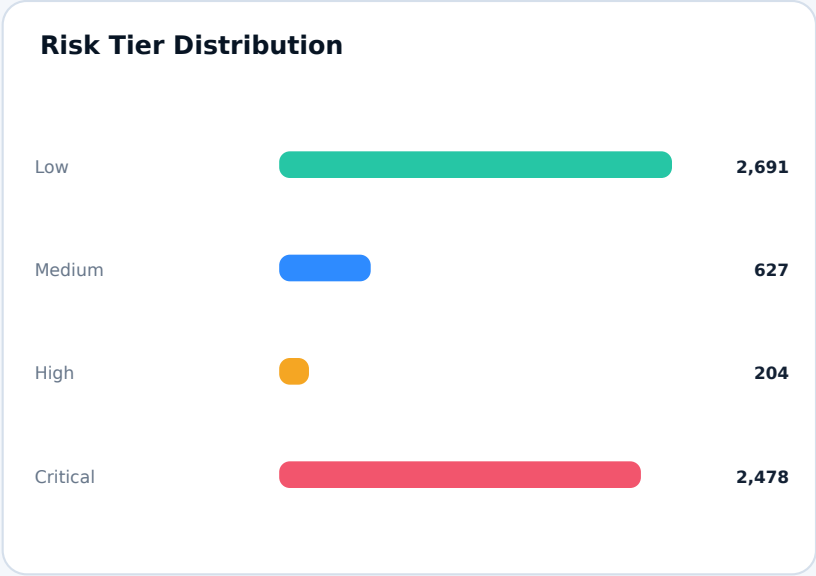
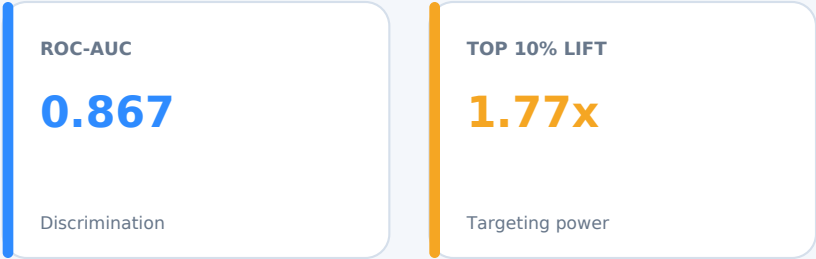
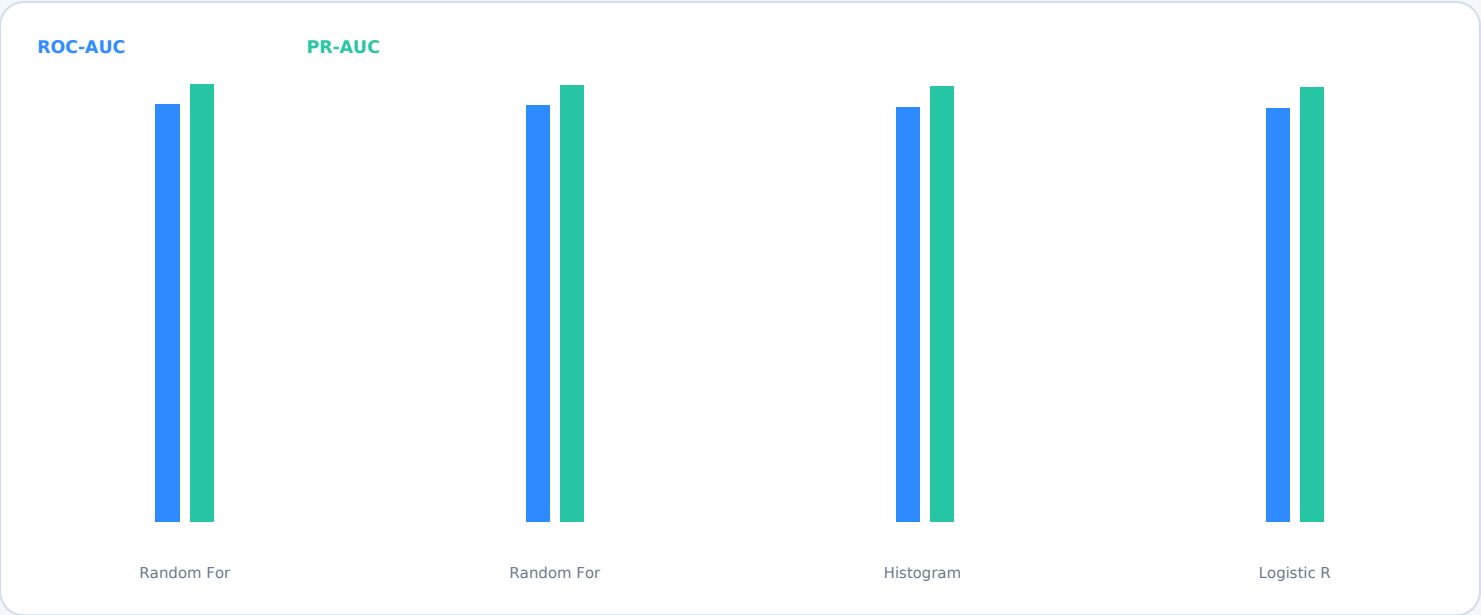
CLV Portfolio by RFM Segment



Sources: clv_model_comparison.csv; clv_predictions.csv; modeling.py

Churn Prediction & Risk Portfolio

Champion selection balances discrimination, calibration and business lift



Leakage control

Training snapshots precede the 31 Dec 2024 holdout; production scoring uses the 31 Dec 2025 portfolio. The operating threshold is selected before the holdout is evaluated.

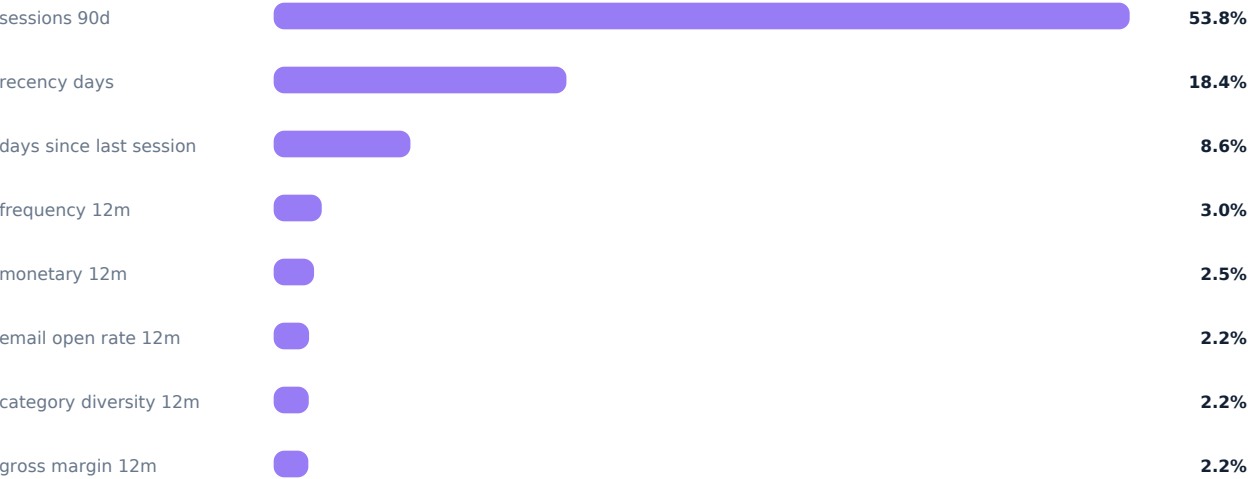
Sources: churn_model_comparison.csv; churn_predictions.csv; model_metadata.json

Synthetic portfolio data • 2022–2025 • Production snapshot: 31 Dec 2025 • TRY

Explainability, Fairness & Drift

Responsible use requires transparent drivers and post-approval controls

Top Global SHAP Drivers



Interpretation boundary

SHAP explains model association, not causality. Human review and campaign eligibility remain above the score.

Region	N	Observed	Avg Risk	AUC
Aegean	621	57.6%	52.0%	0.856
Black Sea	367	54.8%	50.1%	0.885
Central Anatolia	671	57.8%	51.4%	0.873
Eastern Anatolia	159	58.5%	48.1%	0.869
Marmara	1,549	54.2%	50.5%	0.865
Mediterranean	486	56.0%	50.7%	0.871

DRIFT WATCH

tenure days

PSI 0.245

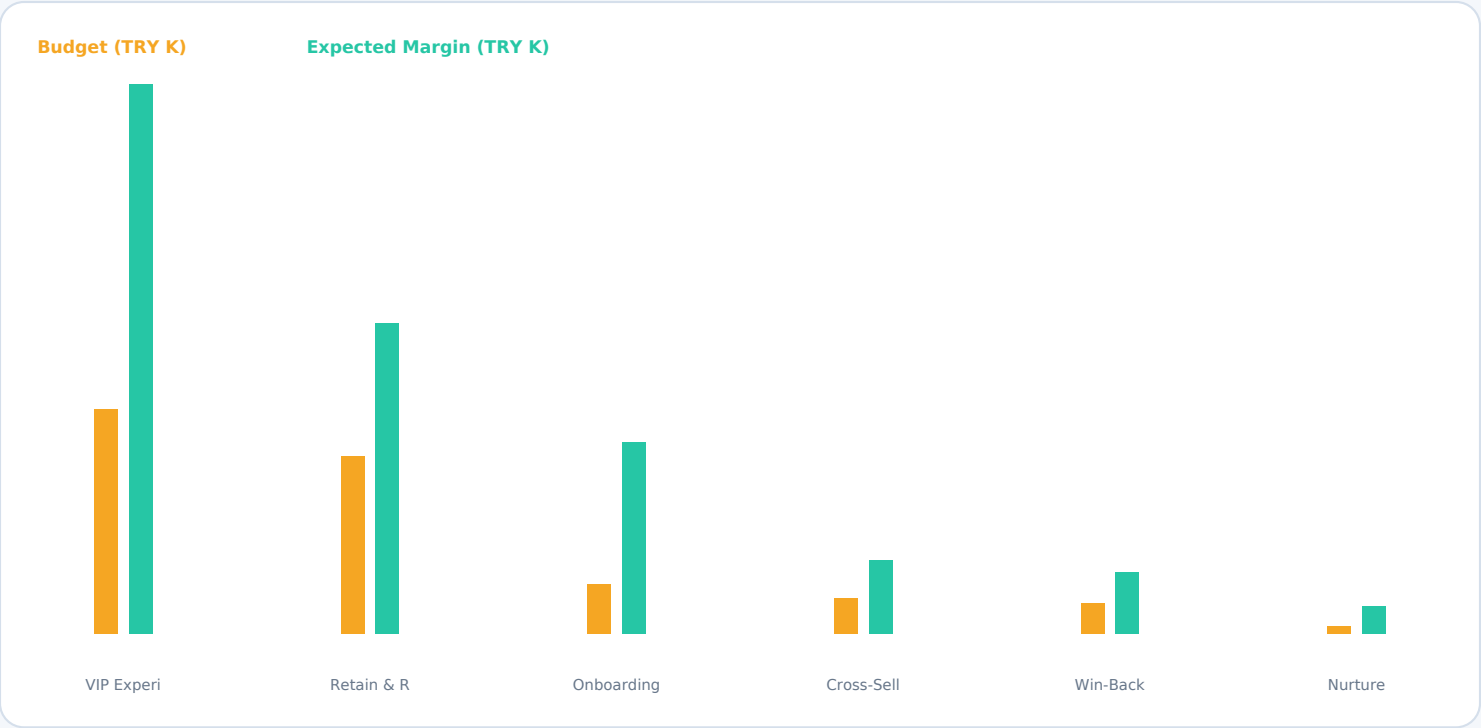
QUALITY GATES

10/10

All pipeline checks PASS

Campaign Targeting & Economics

Budget allocated by expected incremental margin and controlled ROI



Action	Selected	Budget	Margin	ROI
VIP Experience	1,065	TRY 101K	TRY 248K	2.38x
Retain & Reward	1,109	TRY 80K	TRY 140K	1.71x
Onboarding	652	TRY 22K	TRY 86K	3.74x
Cross-Sell	324	TRY 16K	TRY 33K	1.89x
Win-Back	228	TRY 14K	TRY 28K	0.25x
Nurture	123	TRY 3K	TRY 12K	3.54x

PRIORITIZED

3,501

Customer actions

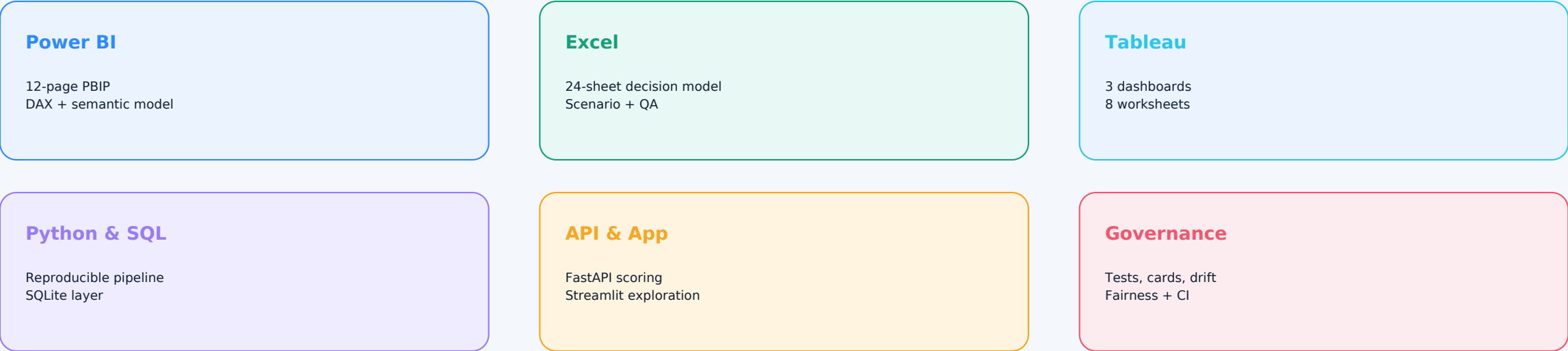
EXPECTED MARGIN

TRY 547K

Incremental contribution

Delivery Architecture & 90-Day Roadmap

From portfolio prototype to an operating customer-intelligence capability



Know customer value. Anticipate risk. Act with discipline.

Sources: README.md; PowerBI; Excel; Tableau; API; Docs

Synthetic portfolio data • 2022–2025 • Production snapshot: 31 Dec 2025 • TRY